

GSEA Null Calibration: Permutation vs Rotation (Walkthrough)

[Portfolio](#)

Overview

This document explains the main steps in `rotations.r`, runs the simulation script, and summarizes the differences between gene-wise permutation and rotation-based null generation.

The report is designed to rerun quickly:

- `rotations.r` now uses a simulation cache (`simulation_cache.rds`) for Monte Carlo outputs
- this Quarto document also enables knitr chunk caching

If the cache does not exist (or configuration changes), the first run can still take a while.

What `rotations.r` Does (Step-by-Step)

1. Configuration

The script sets:

- dataset dimensions (`ngenes`, `nsamples`)
- gene set size (`set_size`, `set_idx`)
- Monte Carlo settings (`nsim`, `nperm`, `nrot`)
- correlation settings (`rho_set_grid`, `rho_bg`)
- parallelization mode (`parallel_over`, `ncores`)
- cache options (`use_simulation_cache`, `simulation_cache_file`)

2. Core helper functions

The script defines functions to:

- compute a GSEA-like enrichment score (`gsea_set_stat`)
- compute empirical right-tail p-values (`empirical_p_right`)
- simulate correlated noise in set and background genes
- simulate null datasets (no differential expression)
- fit limma and compute enrichment scores from p-values
- build a rotation basis from the null-model residual space
- sample random orthogonal rotations in that residual space

3. Null generation methods compared

For each simulated dataset, the script computes:

- **Permutation null:** shuffles gene-wise p-values before recomputing the set statistic
- **Rotation null:** rotates residuals under the null model, refits, and recomputes the set statistic

This is repeated over:

- multiple correlation strengths in `rho_set_grid`
- multiple Monte Carlo replicates (`nsim`)

4. Summaries and output files

The script then:

- summarizes empirical type-I error and mean p-values by `rho_set`
- writes CSV outputs
- creates diagnostic plots (calibration panel, ECDF, null histograms)
- creates correlation heatmaps and rotation-preservation diagnostics

Run the Script (Using Cached Simulation Data)

```

if (!file.exists("rotations.r")) {
  stop("Cannot find rotations.r in the working directory.")
}

if (!file.exists("simulation_cache.rds")) {
  message("No simulation cache found yet. First run may be slow.")
}

sim_env <- new.env(parent = globalenv())
source("rotations.r", local = sim_env)

```

Loaded cached simulation outputs from simulation_cache.rds

Empirical type-I error (target = 0.05):

	rho_set	type1_perm	type1_rot	mean_p_perm	mean_p_rot
1	0.0	0.11	0.04	0.4667864	0.4805788
2	0.2	0.15	0.03	0.4641517	0.4530539
3	0.4	0.23	0.05	0.5284631	0.4839920
4	0.6	0.40	0.05	0.4893413	0.4638523
5	0.8	0.39	0.08	0.5352695	0.5249301

Heatmap summary stats:

	rho_set	mean_within_set	mean_within_bg	mean_between
1	0.0	0.003047067	0.09257803	0.005082836
2	0.2	0.138229143	0.08599253	0.056223204
3	0.4	0.388186175	0.09306683	0.057268731
4	0.6	0.625472965	0.04222763	-0.002154815
5	0.8	0.780083479	0.14768735	-0.087009940

Rotation correlation-preservation summary:

	rho_set	pairwise_corr_of_corr	mean_abs_diff	n_genes_plotted	n_pairs
1	0.0		1 1.751655e-16	400	79800
2	0.8		1 1.504541e-16	400	79800

Saved:

- null_calibration_summary.csv
- null_calibration_results.csv
- null_calibration.pdf

- corr_heatmap_summary.csv
- corr_heatmap_rho_*.pdf
- rotation_corr_preservation_summary.csv
- rotation_corr_preservation_rho_*.pdf
- simulation_cache.rds

```
# Pull key objects into the report environment for convenience
needed <- c(
  "ngenes", "nsamples", "set_size", "set_idx", "nsim", "nperm", "nrot", "alpha",
  "rho_set_grid", "rho_bg", "parallel_over", "ncores",
  "results", "summary_tbl", "example_datasets", "example_distributions",
  "rotation_preservation_summary",
  "build_rotation_basis", "sample_rotation_matrix"
)
for (nm in needed) {
  if (exists(nm, envir = sim_env, inherits = FALSE)) {
    assign(nm, get(nm, envir = sim_env, inherits = FALSE))
  }
}
```

Configuration Used in This Run

```
config_tbl <- data.frame(
  parameter = c(
    "ngenes", "nsamples", "set_size", "nsim", "nperm", "nrot", "alpha",
    "rho_set_grid", "rho_bg", "parallel_over", "ncores"
  ),
  value = c(
    ngenes, nsamples, set_size, nsim, nperm, nrot, alpha,
    paste(rho_set_grid, collapse = ", "),
    rho_bg, parallel_over, ncores
  ),
  stringsAsFactors = FALSE
)
knitr::kable(config_tbl, caption = "Configuration used by rotations.r")
```


Table 1: Configuration used by rotations.r

parameter	value
ngenes	2000
nsamples	12
set_size	100
nsim	100
nperm	500
nrot	500
alpha	0.05
rho_set_grid	0, 0.2, 0.4, 0.6, 0.8
rho_bg	0.1
parallel_over	nsim
ncores	7

Simulation Outputs

Monte Carlo results (head)

```
knitr::kable(head(results, 10), caption = "First 10 Monte Carlo rows")
```

Table 2: First 10 Monte Carlo rows

rho_set	sim	p_perm	p_rot
0	1	0.9720559	0.9660679
0	2	0.5568862	0.6127745
0	3	0.2495010	0.3992016
0	4	0.4051896	0.4471058
0	5	0.3253493	0.3712575
0	6	0.1257485	0.2534930
0	7	0.6307385	0.7285429
0	8	0.5808383	0.4950100
0	9	0.6067864	0.4630739
0	10	0.9540918	0.9960080

Calibration summary by correlation level

```
summary_view <- summary_tbl
summary_view$delta_type1_perm_minus_rot <- summary_view$type1_perm - summary_view$type1_rot
summary_view$delta_mean_p_perm_minus_rot <- summary_view$mean_p_perm - summary_view$mean_p_rot
knitr::kable(summary_view, digits = 4, caption = "Calibration summary across rho_set")
```

Table 3: Calibration summary across rho_set

rho_set	type1_perm	type1_rot	mean_p_perm	mean_p_rot	delta_type1_perm_minus_rot	delta_mean_p_perm_minus_rot
0.0	0.11	0.04	0.4668	0.4806	0.07	-0.0138
0.2	0.15	0.03	0.4642	0.4531	0.12	0.0111
0.4	0.23	0.05	0.5285	0.4840	0.18	0.0445
0.6	0.40	0.05	0.4893	0.4639	0.35	0.0255
0.8	0.39	0.08	0.5353	0.5249	0.31	0.0103

Visualizing the Difference: Permutation vs Rotation

1. Type-I error calibration vs within-set correlation

This reproduces the main “Calibration Under Null” comparison and highlights how the methods behave as gene-gene correlation increases.

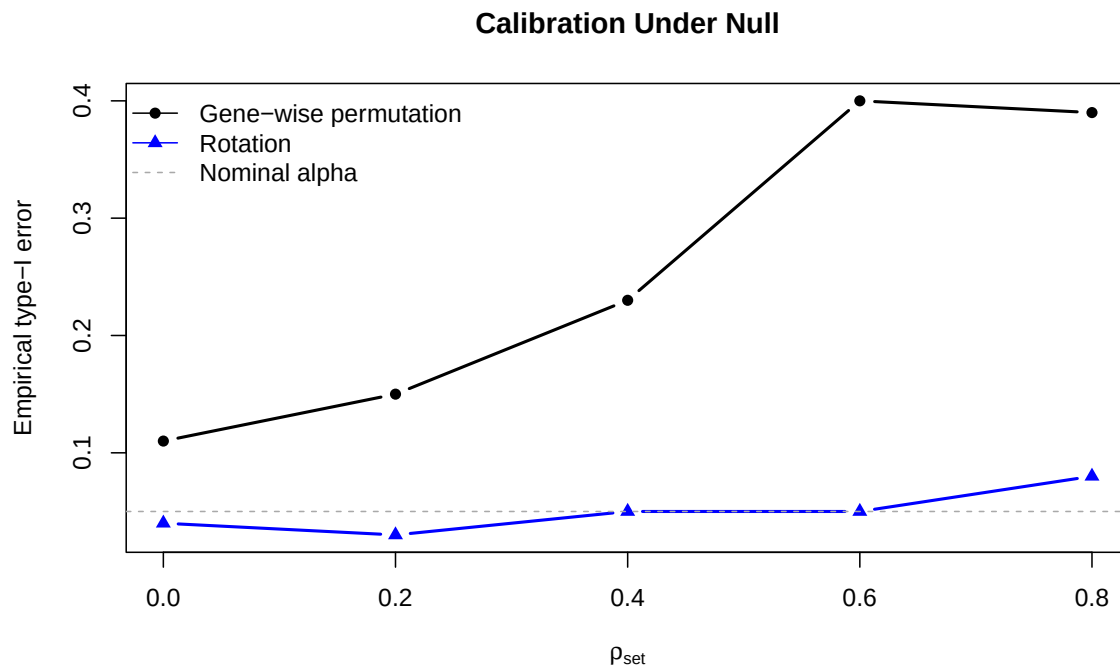
```
plot_tbl <- summary_tbl[
  is.finite(summary_tbl$rho_set) &
  is.finite(summary_tbl$type1_perm) &
  is.finite(summary_tbl$type1_rot),
, drop = FALSE]
plot_tbl <- plot_tbl[order(plot_tbl$rho_set), , drop = FALSE]

if (nrow(plot_tbl) == 0) {
  plot.new()
  title("Calibration Under Null")
  text(0.5, 0.5, "No finite values to plot")
} else {
  y_lim <- range(c(plot_tbl$type1_perm, plot_tbl$type1_rot, alpha), na.rm = TRUE)
  if (y_lim[1] == y_lim[2]) y_lim <- y_lim + c(-0.01, 0.01)
```

```

plot(
  plot_tbl$rho_set, plot_tbl$type1_perm,
  type = "b", pch = 16, lwd = 2,
  ylim = y_lim,
  xlab = expression(rho[set]),
  ylab = "Empirical type-I error",
  main = "Calibration Under Null"
)
lines(plot_tbl$rho_set, plot_tbl$type1_rot, type = "b", pch = 17, lwd = 2, col = "blue")
abline(h = alpha, lty = 2, col = "darkgray")
legend(
  "topleft",
  legend = c("Gene-wise permutation", "Rotation", "Nominal alpha"),
  col = c("black", "blue", "darkgray"),
  lty = c(1, 1, 2),
  pch = c(16, 17, NA),
  bty = "n"
)
}

```

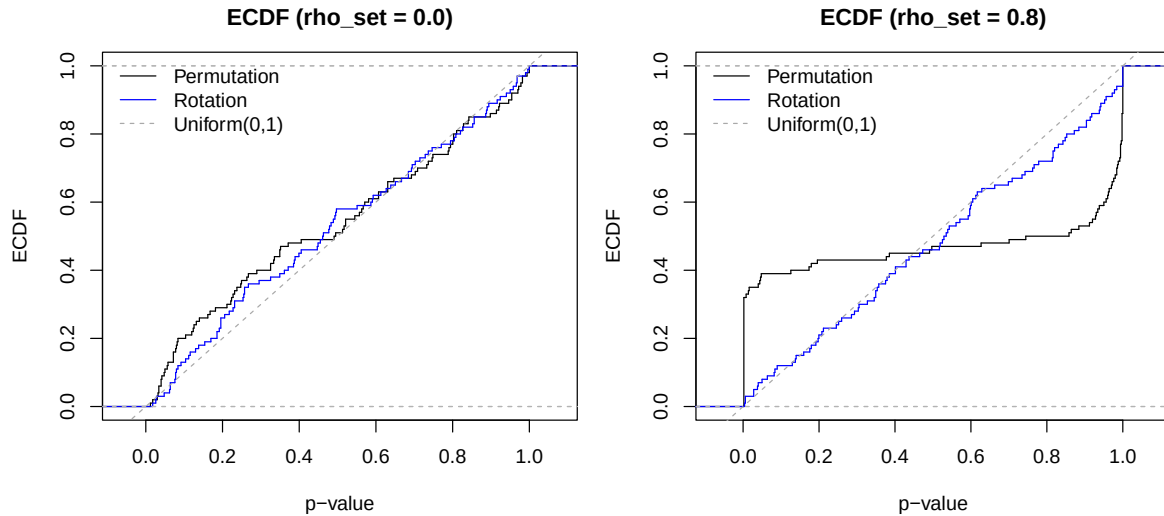


2. p-value ECDF comparison at low and high correlation

Uniformity under the null is a useful diagnostic. The dashed line is the ideal uniform ECDF.

```
par(mfrow = c(1, 2), mar = c(4, 4, 3, 1))

for (rho_val in c(min(rho_set_grid), max(rho_set_grid))) {
  idx <- results$rho_set == rho_val & is.finite(results$p_perm) & is.finite(results$p_rot)
  if (!any(idx)) {
    plot.new()
    title(sprintf("ECDF (rho_set = %.1f)", rho_val))
    text(0.5, 0.5, "No finite p-values")
  } else {
    plot(
      ecdf(results$p_perm[idx]),
      do.points = FALSE,
      verticals = TRUE,
      main = sprintf("ECDF (rho_set = %.1f)", rho_val),
      xlab = "p-value",
      ylab = "ECDF"
    )
    lines(ecdf(results$p_rot[idx]), do.points = FALSE, verticals = TRUE, col = "blue")
    abline(0, 1, lty = 2, col = "darkgray")
    legend(
      "topleft",
      legend = c("Permutation", "Rotation", "Uniform(0,1)"),
      col = c("black", "blue", "darkgray"),
      lty = c(1, 1, 2),
      bty = "n"
    )
  }
}
```



```
par(mfrow = c(1, 1))
```

3. Example null distributions of the enrichment statistic

These panels compare the null distributions produced by the two methods for a single example dataset at low and high correlation.

```
par(mfrow = c(1, 2), mar = c(4, 4, 3, 1))

plot_example_null <- function(ex, rho_label) {
  if (is.null(ex)) {
    plot.new()
    title(sprintf("Example null ES (rho_set = %s)", rho_label))
    text(0.5, 0.5, "Not available")
    return(invisible(NULL))
  }
  hist(
    ex$null_perm, breaks = 40, col = "gray85", border = "white",
    main = sprintf("Example null ES (rho_set = %s)", rho_label),
    xlab = "ES"
  )
  hist(ex$null_rot, breaks = 40, col = rgb(0, 0, 1, 0.35), border = "white", add = TRUE)
  abline(v = ex$ES_obs, col = "red", lwd = 2)
  legend(
    "topright",
```

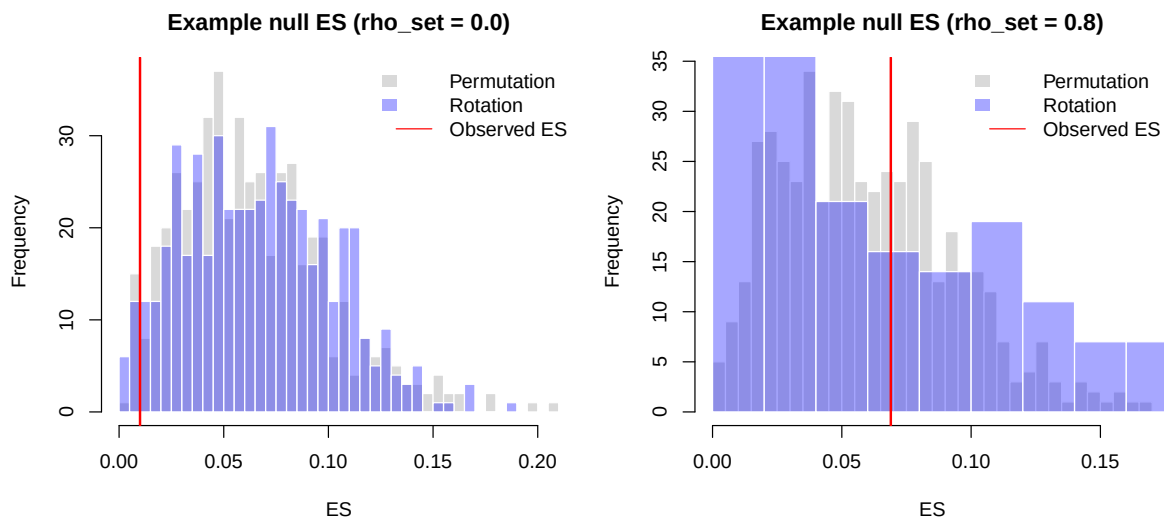
```

    legend = c("Permutation", "Rotation", "Observed ES"),
    fill = c("gray85", rgb(0, 0, 1, 0.35), NA),
    border = NA,
    lty = c(NA, NA, 1),
    col = c(NA, NA, "red"),
    bty = "n"
  )
}

ex_low <- example_distributions[[paste0("rho_", min(rho_set_grid))]]
ex_high <- example_distributions[[paste0("rho_", max(rho_set_grid))]]

plot_example_null(ex_low, sprintf("%.1f", min(rho_set_grid)))
plot_example_null(ex_high, sprintf("%.1f", max(rho_set_grid)))

```



```

par(mfrow = c(1, 1))

```

Why Rotation Helps: Correlation Structure Preservation

The key idea is that rotation operates in the residual space and preserves the covariance/correlation structure more faithfully than gene-wise permutation.

4. Correlation heatmaps from the simulated datasets

These are the summary statistics generated by `rotations.r` from example datasets.

```
if (file.exists("corr_heatmap_summary.csv")) {  
  corr_heatmap_summary <- read.csv("corr_heatmap_summary.csv")  
  knitr::kable(corr_heatmap_summary, digits = 4, caption = "Correlation heatmap summary statistics")  
} else {  
  cat("`corr_heatmap_summary.csv` not found. Re-run `rotations.r`.\n")  
}
```

Table 4: Correlation heatmap summary statistics

rho_set	mean_within_set	mean_within_bg	mean_between
0.0	0.0030	0.0926	0.0051
0.2	0.1382	0.0860	0.0562
0.4	0.3882	0.0931	0.0573
0.6	0.6255	0.0422	-0.0022
0.8	0.7801	0.1477	-0.0870

5. Rotation correlation-preservation metrics

This directly quantifies how similar the pairwise gene-gene correlations are before vs after a residual-space rotation.

```
if (exists("rotation_preservation_summary")) {  
  knitr::kable(rotation_preservation_summary, digits = 4, caption = "Rotation correlation-preservation summary")  
} else if (file.exists("rotation_corr_preservation_summary.csv")) {  
  rotation_preservation_summary <- read.csv("rotation_corr_preservation_summary.csv")  
  knitr::kable(rotation_preservation_summary, digits = 4, caption = "Rotation correlation-preservation summary")  
} else {  
  cat("Rotation preservation summary not available.\n")  
}
```

Table 5: Rotation correlation-preservation summary

rho_set	pairwise_corr_of_corr	mean_abs_diff	n_genes_plotted	n_pairs
0.0	1	0	400	79800
0.8	1	0	400	79800

6. Inline correlation-preservation visualization (original vs rotated)

The next chunk recreates the key idea visually for the lowest and highest `rho_set` example datasets:

- original gene-gene correlation heatmap
- rotated gene-gene correlation heatmap
- scatter of pairwise correlations (original vs rotated)

```
plot_rotation_preservation_inline <- function(Y, set_idx, rho_val, bg_n = 300) {  
  if (is.null(Y)) {  
    plot.new()  
    title(sprintf("rho_set = %.1f", rho_val))  
    text(0.5, 0.5, "Example dataset not available")  
    return(invisible(NULL))  
  }  
  
  nsamp <- ncol(Y)  
  design0 <- matrix(1, nrow = nsamp, ncol = 1)  
  fit0 <- limma::lmFit(Y, design0)  
  Yhat0 <- fitted(fit0)  
  res0 <- Y - Yhat0  
  basis <- build_rotation_basis(design0 = design0, nsamples = nsamp)  
  Rmat <- sample_rotation_matrix(basis)  
  Y_rot <- Yhat0 + res0 %*% Rmat  
  
  bg_idx <- setdiff(seq_len(nrow(Y)), set_idx)  
  if (length(bg_idx) > bg_n) bg_idx <- sample(bg_idx, bg_n)  
  keep_idx <- c(set_idx, bg_idx)  
  
  corr_orig <- cor(t(Y[keep_idx, , drop = FALSE]))  
  corr_rot <- cor(t(Y_rot[keep_idx, , drop = FALSE]))  
  u <- upper.tri(corr_orig)  
  x <- corr_orig[u]  
  y <- corr_rot[u]  
  r_xy <- suppressWarnings(cor(x, y, use = "complete.obs"))  
  mad_xy <- mean(abs(x - y), na.rm = TRUE)  
  
  cols <- colorRampPalette(c("#313695", "#ffffbf", "#a50026"))(100)  
  
  image(  
    seq_len(nrow(corr_orig)), seq_len(ncol(corr_orig)),  
    corr_orig[nrow(corr_orig):1, ],  
    cols, mad_xy)
```



```

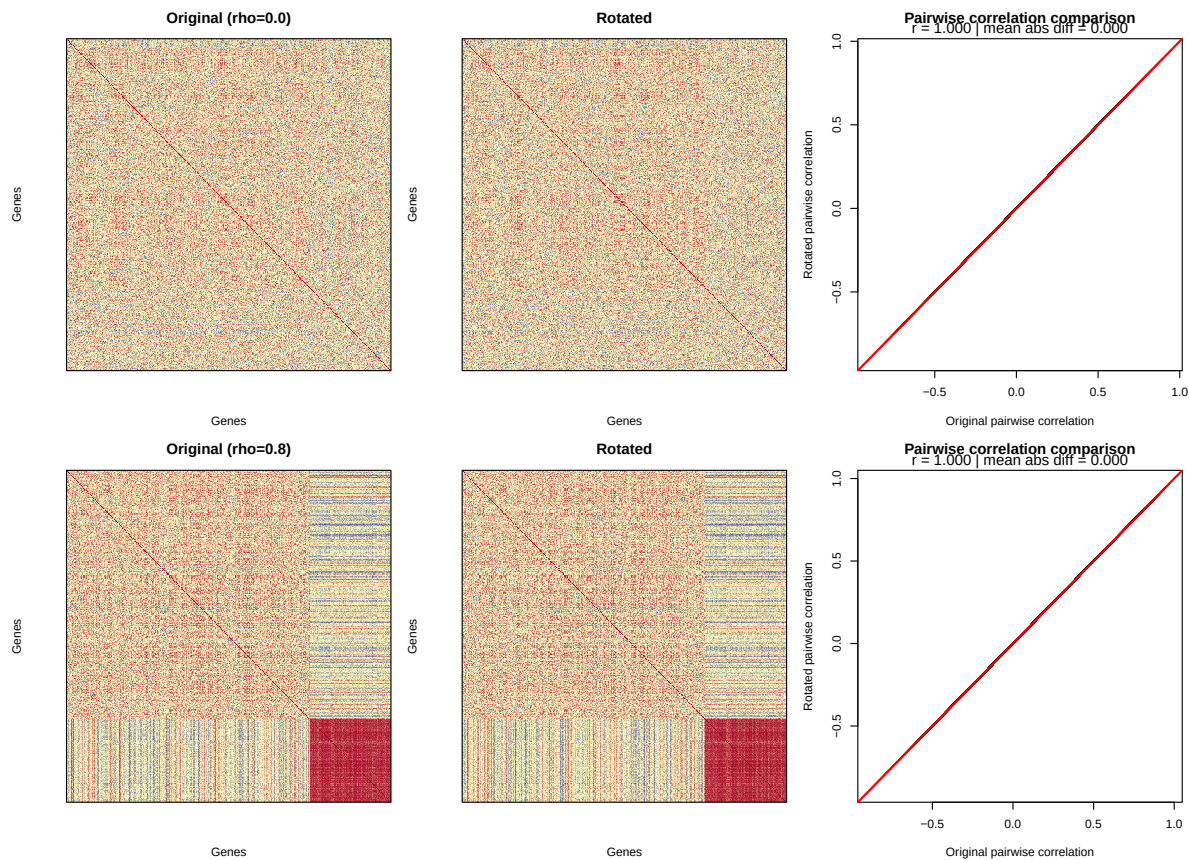
    col = cols, zlim = c(-1, 1), axes = FALSE,
    xlab = "Genes", ylab = "Genes",
    main = sprintf("Original (rho=%.1f)", rho_val)
)
box()

image(
  seq_len(nrow(corr_rot)), seq_len(ncol(corr_rot)),
  corr_rot[nrow(corr_rot):1, ],
  col = cols, zlim = c(-1, 1), axes = FALSE,
  xlab = "Genes", ylab = "Genes",
  main = "Rotated"
)
box()

plot(
  x, y,
  pch = 16, cex = 0.35,
  col = rgb(0, 0, 0, 0.18),
  xlab = "Original pairwise correlation",
  ylab = "Rotated pairwise correlation",
  main = "Pairwise correlation comparison"
)
abline(0, 1, col = "red", lwd = 2)
mtext(sprintf("r = %.3f | mean abs diff = %.3f", r_xy, mad_xy), side = 3, line = 0.2, cex = 0.8)
}

par(mfrow = c(2, 3), mar = c(4, 4, 3, 1))
plot_rotation_preservation_inline(example_datasets[[paste0("rho_", min(rho_set_grid))]], set.seed(1))
plot_rotation_preservation_inline(example_datasets[[paste0("rho_", max(rho_set_grid))]], set.seed(1))

```



```
par(mfrow = c(1, 1))
```

Interpretation Guide

When you read the report, the main differences to focus on are:

- **Type-I error calibration:**
 - if permutation inflates type-I error as **rho_set** increases, it suggests correlation is not being respected by the null
 - rotation should remain closer to nominal **alpha**
- **p-value ECDFs:**
 - departures from the diagonal under the null indicate miscalibration
- **Null ES histograms:**
 - compare how the null distributions differ in spread/shape

- **Correlation-preservation plots:**
 - points near the $y = x$ line indicate rotated data preserve pairwise correlations well

Files Produced by `rotations.r`

```
produced_files <- c(
  "simulation_cache.rds",
  "null_calibration_summary.csv",
  "null_calibration_results.csv",
  "null_calibration.pdf",
  "corr_heatmap_summary.csv",
  "rotation_corr_preservation_summary.csv"
)

file_status <- data.frame(
  file = produced_files,
  exists = file.exists(produced_files),
  stringsAsFactors = FALSE
)

knitr::kable(file_status, caption = "Key output files")
```

Table 6: Key output files

file	exists
simulation_cache.rds	TRUE
null_calibration_summary.csv	TRUE
null_calibration_results.csv	TRUE
null_calibration.pdf	TRUE
corr_heatmap_summary.csv	TRUE
rotation_corr_preservation_summary.csv	TRUE

```
cat("\nHeatmap PDFs:", length(Sys.glob("corr_heatmap_rho_*.pdf")), "\n")
```

Heatmap PDFs: 5

```
cat("Rotation-preservation PDFs:", length(Sys.glob("rotation_corr_preservation_rho_*.pdf")),
```

Rotation-preservation PDFs: 2